$$i\frac{\partial U}{\partial z} = \frac{\beta_2}{2}\frac{\partial^2 U}{\partial t^2} + \frac{i\beta_3}{6}\frac{\partial^3 U}{\partial t^3}$$

而可由 FT 求解：

$$U(z,T) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{U}(0,\omega)\exp\left(\frac{i}{2}\beta_2\omega^2 z + \frac{i}{6}\beta_3\omega^3 z - i\omega T\right)d\omega.$$

$$\tilde{U}(0,\omega) = \int_{-\infty}^{\infty} U(0,T)\, e^{i\omega T}\, dT$$

对 chirped Gaussian pulse，

$$\tilde{U}(0,\omega) = \left(\frac{2\pi T_0^2}{1+iC}\right)^{1/2}\exp\left[-\frac{\omega^2 T_0^2}{2(1+iC)}\right]$$

引入 $x \equiv \omega p$，$d\omega = \frac{1}{p}dx$

$$p^2 = \frac{T_0^2}{2}\left(\frac{1}{1+iC} - \frac{i\beta_2 z}{T_0^2}\right).$$

有：

$$U(z,T) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{2\pi T_0^2}{1+iC}\right)^{1/2}\exp\left[-\frac{\omega^2 T_0^2}{2(1+iC)}\right]\exp\left(\frac{i}{2}\beta_2\omega^2 z + \frac{i}{6}\beta_3\omega^3 z - i\omega T\right)d\omega.$$

$$= \frac{A_0}{\sqrt{\pi}}\int_{-\infty}^{\infty}\exp\left(-x^2 + \frac{ib}{3}x^3 - \frac{iT}{p}x\right)dx.$$

其中，$b \equiv \beta_3 z/(2p^3)$。$x^2$ 项可用另一变换 $x = b^{-1/3}u - i/b$ 消去：

最后为艾里函数：

$$U(z,T) = \frac{2A_0\sqrt{\pi}}{|b|^{1/3}}\exp\left(\frac{2p-3bT}{3pb^2}\right)Ai\left(\frac{p-bT}{p|b|^{1/3}}\right)$$

引入 TOD 长度

$$L_D' \equiv T_0^3/|\beta_3|$$

only when $L_D' \leq L_D$ or $T_0 |\beta_2/\beta_1| \leq 1$, TOD 效应明显.

· TOD 使脉冲形状畸变并产生长尾的拖尾.

展宽因子?

$|U(z,T)|^2$ 作 FT.

$$\tilde{I}(z,\omega) = \int_{-\infty}^{\infty} |U(z,T)|^2 \exp(i\omega T)\, dT.$$

n次微分:

$$\lim_{\omega\to 0} \frac{\partial^n}{\partial \omega^n} \tilde{I}(z,\omega) = (i)^n \int_{-\infty}^{\infty} T^n |U(z,T)|^2 dT.$$

T 的 n 阶矩

$$\langle T^n \rangle = \frac{\int_{-\infty}^{\infty} T^n |U(z,T)|^2 dT}{\int_{-\infty}^{\infty} |U(z,T)|^2 dT} = \frac{\frac{1}{i^n} \lim_{\omega\to 0} \frac{\partial^n}{\partial \omega^n} \tilde{I}(z,\omega)}{\int_{-\infty}^{\infty} |U(z,T)|^2 dT.} = \frac{(-i)^n}{N_c} \lim_{\omega\to 0} \frac{\partial^n}{\partial \omega^n} \tilde{I}(z,\omega)$$

归一化常数 $N_c \equiv \int_{-\infty}^{\infty} |U(z,T)|^2 dT$.

由卷积定理: ? 时域 '×' $\xrightarrow{FT}$ 频域 '*'.

$$\tilde{I}(z,\omega) = F\{ U(z,T)\, U^*(z,T) \} = \tilde{U}(z,\omega) * \tilde{U}^*(z,\omega)$$

$$= \int_{-\infty}^{\infty} \tilde{U}(z,\omega-\omega')\, \tilde{U}^*(z,\omega')\, d\omega'$$

代入 $\langle T^n \rangle$, 有: $\langle T^n \rangle = \frac{(-i)^n}{N_c} \lim_{\omega\to 0} \frac{\partial^n}{\partial \omega^n} \int_{-\infty}^{\infty} \tilde{U}(z,\omega-\omega')\, \tilde{U}^*(z,\omega')\, d\omega'$.

$\tilde{U}^*(z,\omega')$ 与 ω 无关.

$$\langle T^n \rangle = \frac{i^n}{N_c} \int_{-\omega}^{\omega} \tilde{U}^*(z,\omega) \lim_{\omega \to 0} \frac{\partial^n}{\partial \omega^n} \tilde{U}(z,-\omega)\, d\omega$$

- For Gaussian pulse:

$$\tilde{U}(z,\omega) = \left(\frac{2\pi T_0^2}{1+iC}\right)^{1/2} \exp\left[\frac{i\omega^2}{2}\left(\beta_2 z + \frac{i T_0^2}{1+iC}\right) + \frac{i}{6}\beta_3 \omega^3 z\right]$$

脉冲的 RMS 宽度：

$$\sigma = \left[\langle T^2 \rangle - \langle T \rangle^2\right]^{1/2}$$

展宽因子：

$$\frac{\sigma}{\sigma_0} = \left[\left(1 + \frac{C\beta_2 z}{2\sigma_0^2}\right)^2 + \left(\frac{\beta_2 z}{2\sigma_0^2}\right)^2 + (1+C^2)^2 \frac{1}{2}\left(\frac{\beta_2 z}{4\sigma_0^2}\right)^2\right]^{1/2}$$

其中，σ_0 为啁啾高斯脉冲初始均方根宽度。$\sigma_0 = \frac{T_0}{\sqrt{2}}$